Information gain: 2 practice problems

Embedded AI midterm preparation, level of the practice series (practice 2, problem 3)

Hint (printed on the practice sheets):

$$\text{IGain}(T) = I\Big(\tfrac{p}{p+n}, \tfrac{n}{p+n}\Big) - \sum_{k=1}^{K} \frac{p_k + n_k}{p+n}\, I\Big(\tfrac{p_k}{p_k+n_k}, \tfrac{n_k}{p_k+n_k}\Big), \qquad I(a,b) = -a\log_2 a - b\log_2 b$$

Calculator: $\log_2 x = \ln x / \ln 2$.

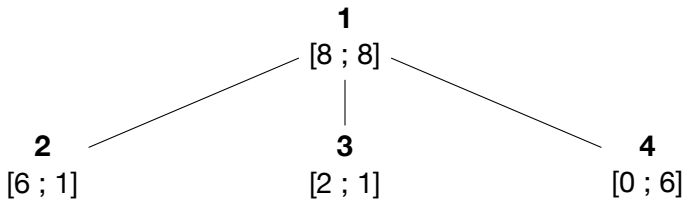
Problem 1. We build a decision tree that predicts whether a sensor node fails during the day. Each situation is described by two attributes:

A. Battery level: Low (L), Medium (M), High (H)
B. Temperature: Hot (Ho), Cold (Co)

Ex.	Battery	Temp.	Failed?	Ex.	Battery	Temp.	Failed?
1	L	Ho	Y	7	L	Ho	N
2	M	Co	Y	8	M	Ho	Y
3	H	Ho	N	9	H	Ho	N
4	L	Co	Y	10	L	Co	Y
5	M	Ho	N	11	M	Co	N
6	H	Co	N	12	H	Co	N

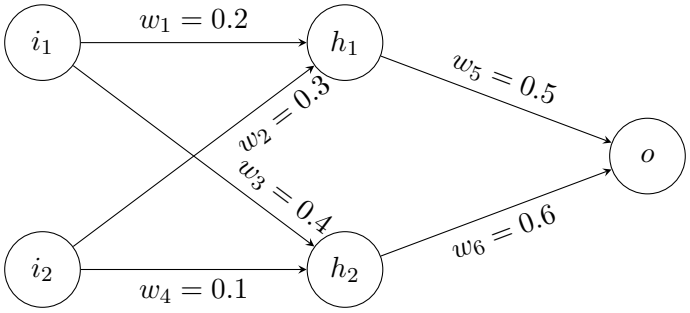
We test the battery level at the root node. Draw the decision tree resulting from this test and calculate the information gain of the test based on the 12 training samples. (Temp. = temperature, Failed? = did the node fail?)

Problem 2. We train the decision tree below for a binary decision. Every example belongs to class A or class B; there were 8 training examples of each class. The numbers in brackets show how many examples of class A and of class B reached each node (class A first). What is the information gain of the test performed at node 1? (Justify with a calculation.)



Problem 3. The network below has 2 inputs, 2 hidden neurons and 1 output neuron. All neurons use the sigmoid activation $f(x) = 1/(1 + e^{-x})$. The hidden neurons share the bias $b_1 = 0.3$, the output neuron has the bias $b_2 = 0.2$. The error is $E = \frac{1}{2}(t - y)^2$.

Input: $i_1 = 0.1$, $i_2 = 0.5$. Desired output (target): $t = 0.8$. Learning factor: $\eta = 0.5$.



a) Perform the forward pass: calculate the outputs of h_1, h_2 and o.

b) Calculate the value of the weight w_5 after the training step.

c) Calculate the value of the weight w_1 after the training step.